

Capstone Project (Basic MoCo I)

Project 1:

1. You have a collection of [500 natural products in SMI](#) compressed file
2. Decompress using Open Babel or RDKit
3. Perform Lipinski Filtering
4. Dock the ligands in the active site of [9NSR](#) (bifunctional farnesyl/geranylgeranyl pyrophosphate synthase (FPPS/GGPPS) from Plasmodium falciparum in complex with MMV019313) using Vina ([box parameters](#) will be provided)
5. Docking Validation (dock redock using the co-crystal ligand and pose reproduction)
6. Visualizations using pymol2 on command line or as a GUI

Project 2:

1. You are provided with a protein ([9NSR](#)), download the protein from RCSB; model it using [AlphaFold](#) from primary sequence; model using [Swiss model](#) from homology models.
2. Fix the missing residues in the 9NSR downloaded from RCSB, using [PDB-Fixer](#). The fixed protein (named **fixed_RCSB_protein**) is [here](#).
3. Go to Enamine database and download a few thousand synthetic compounds from the [PSL-5760 dataset](#) of the diversity library
4. Visualize the chemical space and perform Lipinski filtering
5. Perform clustering and cluster into 10 groups and sample 10 compounds from each group.
6. Dock these 100 compounds against the three ensembles (**fixed_RCSB_protein**, **AlphaFold protein** and **Swiss Model Protein** using vina).
7. Which ensemble gave better affinity
8. Perform S230T mutation on all three ensembles and repeat the docking.
9. Does the mutation affect the docking scores and ranking

Project 3 (Currently Unavailable)

1. You are given a protein (9NSR), download from RCSB
2. Visualize and identify key interactions useful for building a pharmacophore model using ChimeraX.
3. Use Zinc Pharmer to define these constraints and do a pharmacophore search
4. Dock these compounds (**using Pyrex**) and select the best according to docking score.